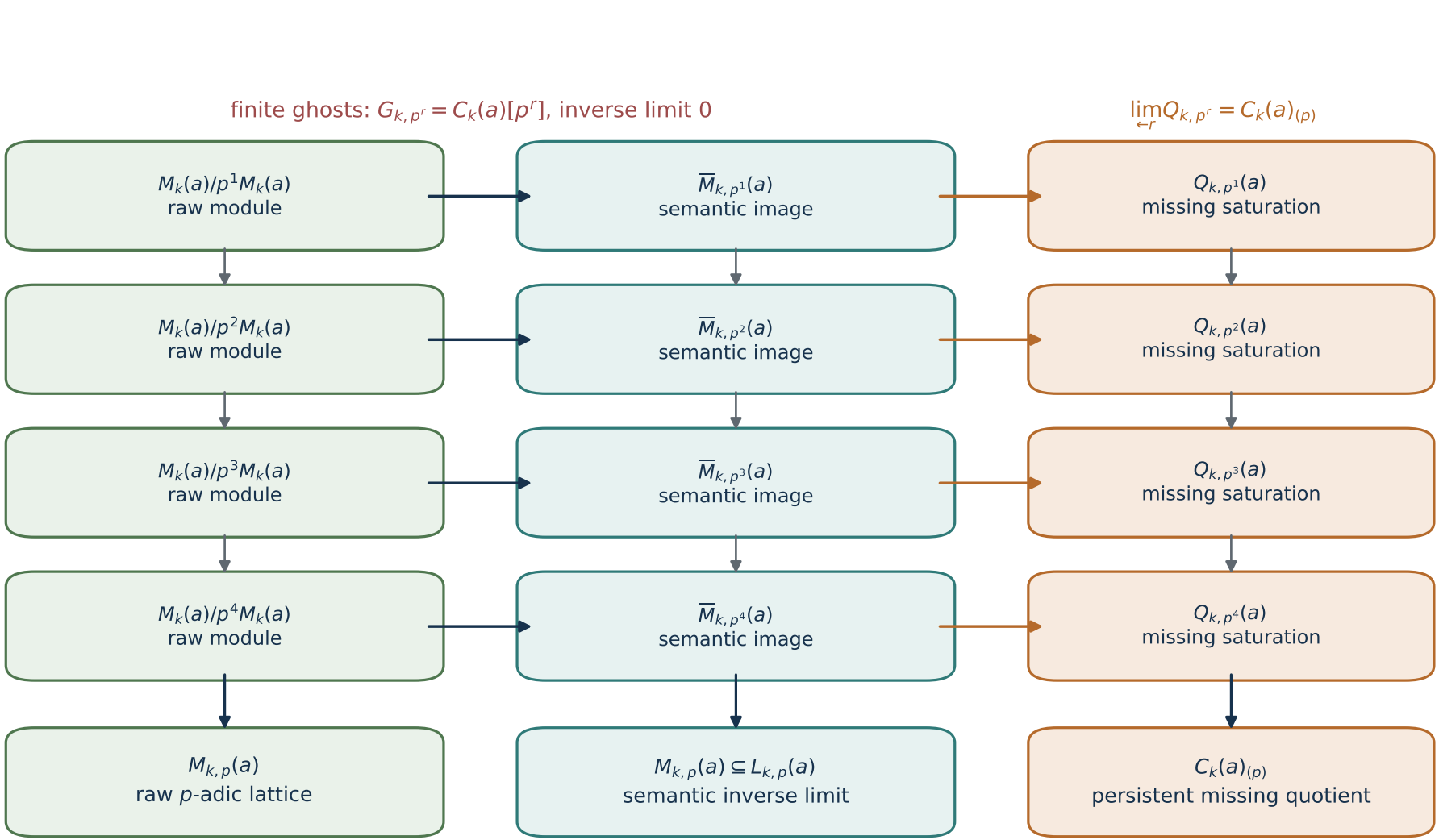
Compatible prime-power towers: raw residues, semantic images, and completion
finite ghosts: $G_{k,p^r} = C_k(a)[p^r]$, inverse limit 0
$\lim_{\leftarrow r} Q_{k,p^r} = C_k(a)_{(p)}$
$M_k(a)/p^1M_k(a)$ raw module
$\overline{M}_{k,p^1}(a)$ semantic image
$Q_{k,p^1}(a)$ missing saturation
$M_k(a)/p^2M_k(a)$ raw module
$\overline{M}_{k,p^2}(a)$ semantic image
$Q_{k,p^2}(a)$ missing saturation
$M_k(a)/p^3M_k(a)$ raw module
$\overline{M}_{k,p^3}(a)$ semantic image
$Q_{k,p^3}(a)$ missing saturation
$M_k(a)/p^4M_k(a)$ raw module
$\overline{M}_{k,p^4}(a)$ semantic image
$Q_{k,p^4}(a)$ missing saturation
$M_{k,p}(a)$ raw p-adic lattice
$M_{k,p}(a) \subseteq L_{k,p}(a)$ semantic inverse limit
$C_k(a)_{(p)}$ persistent missing quotient